

Lecomte

Antoine

Exercise 2 – Network Policies

L'application Online Boutique est une démo d'une architecture microservices composée de plusieurs services frontend et backend qui communiquent entre eux pour simuler une boutique en ligne complète : panier, paiement, catalogue, livraison, email, etc.

Les services principaux sont :

frontend → UI web

checkoutservice → contrôle le checkout

cartservice → gère le panier, stockage du contenu dans Redis

redis-cart → cache Redis pour cartservice

productcatalogservice → donne la liste des produits

currencyservice → convertit les monnaies

paymentservice → simule les paiements

shippingservice → simule les frais de livraison

emailservice → envoie l'email de confirmation

Les services communiquent entre eux avec gRPC ou HTTP.

L'application a été déployée sur un cluster Minikube en suivant les instructions du README, en utilisant les manifests Kubernetes fournis par le dépôt.

Lors du déploiement initial, plusieurs pods n'ont pas pu démarrer à cause d'erreurs ImagePullBackOff. La première tentative de déploiement utilisant "kubectl apply -f" a échoué sur le fichier kustomization. Le problème a été corrigé en déployant les manifests avec "kubectl apply -k" pour traiter correctement les configurations.

Aussi, les manifests Kubernetes fournis faisaient référence à des images de conteneur (comme image: frontend) qui ne sont pas disponibles dans les registres publics. Dans le projet original, ces images sont récupérées via Skaffold ou des composants Kustomize qui n'étaient pas présents dans notre contexte de déploiement.

Correction : nous référençons explicitement les images officielles hébergées sur gcr.io pour chaque service.

Nous avons créé un Pod que nous avons lancé pour la Custom Resource Definition du réseau pour les flux que nous souhaitons explicitement autoriser, et l'annoncer au contrôleur KubeSonde. Ce qu'il doit tester :

```
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl get kubesonde online-boutique-connectivity -o yaml
apiVersion: security.kubesonde.io/v1
kind: Kubesonde
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"security.kubesonde.io/v1","kind":"Kubesonde","metadata":{"annotations":{},"name":"online-boutique-connectivity","namespace":"default"},"spec":{"include":[{"expected":"allow","fromPodSelector":{"app":"frontend","port":"5050","protocol":"TCP","toPodSelector":{"app":"checkoutservice"}},{"expected":"deny","fromPodSelector":{"app":"frontend","port":"6379","protocol":"TCP","toPodSelector":{"app":"redis-cart"}},{"expected":"allow","fromPodSelector":{"app":"cartservice","port":"6379","protocol":"TCP","toPodSelector":{"app":"redis-cart"}},{"expected":"deny","fromPodSelector":{"app":"cartservice","port":"50051","protocol":"TCP","toPodSelector":{"app":"paymentservice"}}],"namespace":"default","probe":"all"}}
  creationTimestamp: "2025-12-24T11:31:33Z"
  generation: 1
  name: online-boutique-connectivity
  namespace: default
  resourceVersion: "23277"
  uid: 9c50f899-1576-4f3e-9a6f-a7fbb2963785
spec:
  include:
  - expected: allow
    fromPodSelector: app=frontend
    port: "5050"
    protocol: TCP
    toPodSelector: app=checkoutservice
  - expected: deny
    fromPodSelector: app=frontend
    port: "6379"
    protocol: TCP
    toPodSelector: app=redis-cart
  - expected: allow
    fromPodSelector: app=cartservice
    port: "6379"
    protocol: TCP
    toPodSelector: app=redis-cart
  - expected: deny
    fromPodSelector: app=cartservice
    port: "50051"
    protocol: TCP
    toPodSelector: app=paymentservice
  namespace: default
  probe: all
user@deb:~/Documents/exam/microservices-demo/network-policies$
```

Nous avons commencé par observer le comportement par défaut de Kubernetes en l'absence de NetworkPolicies (version 0). Tout le trafic réseau entre les pods est autorisé et la configuration est évidemment trop permissive : elle ne respecte pas les exigences de sécurité car KubeSonde détecte des accès interdits qui passent.

```
2025-12-24T12:36:57Z      INFO      kubsonde.podEvents      Active pods: [checkoutservice-7d7bb99d78-4cbvv currencyservic
e-5f49c7988-qwnl9 emailservice-68cd6548d4-ws9hs frontend-56db6cc6df-7wqrt loadgenerator-59ccddd5c-5rnq9 recommendatio
nservice-5856f9b4bb-5tr9s adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 paymentservice-845fc4d44-pd2mq prod
uctcatalogservice-6c6c594f5d-c9sv4 redis-cart-c8ff86559-7vwq5 shippingservice-6d787df574-ggjd5]
2025-12-24T12:36:58Z      INFO      kubsonde.podEvents      Number of probes in State: 533
2025-12-24T12:37:02Z      INFO      Kubsonde Runner        Error when Probing: command terminated with exit code 1
2025-12-24T12:37:04Z      INFO      Kubsonde Runner        Error when Probing: command terminated with exit code 1
2025-12-24T12:37:05Z      INFO      Kubsonde Runner        Error when Probing: command terminated with exit code 1
2025-12-24T12:37:06Z      INFO      Kubsonde Runner        Error when Probing: command terminated with exit code 1
2025-12-24T12:37:07Z      INFO      Kubsonde Runner        Error when Probing: command terminated with exit code 1
2025-12-24T12:37:10Z      INFO      Kubsonde Runner        Error when Probing: command terminated with exit code 1
2025-12-24T12:37:12Z      INFO      Kubsonde Runner        Error when Probing: command terminated with exit code 1
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl logs -n kubsonde-system deploy/kubsonde-cont
roller-manager | grep Probe
2025-12-24T13:09:50Z      INFO      Recursive probing      Probe queue size: 0
2025-12-24T13:12:07Z      INFO      Recursive probing      Probe queue size: 0
2025-12-24T13:13:06Z      INFO      Recursive probing      Probe queue size: 0
2025-12-24T13:14:11Z      INFO      Recursive probing      Probe queue size: 0
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl logs -n kubsonde-system deploy/kubsonde-cont
roller-manager | grep expected
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl logs -n kubsonde-system deploy/kubsonde-cont
roller-manager | grep frontend
2025-12-24T13:09:21Z      INFO      kubsonde.podEvents      EventHandler::AddServiceEvent frontend
2025-12-24T13:09:21Z      INFO      kubsonde.podEvents      EventHandler::AddServiceEvent frontend-external
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl get networkpolicy
No resources found in default namespace.
```

Dans la version 1, deny-all, il n'y a plus aucune communication entre pods, il y a toujours les erreurs de probing "command terminated with exit code 1", mais parce que cette fois, des accès autorisés sont bloqués.

```
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl apply -f v1-deny-all/deny-all.yaml
networkpolicy.networking.k8s.io/deny-all created
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl get networkpolicy
NAME          POD-SELECTOR  AGE
deny-all     <none>        35s
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl logs -n kubernetes-system deploy/kubedns-controller-manager
2025-12-24T13:40:05Z      INFO      setup      starting Kubedns API server
2025-12-24T13:40:05Z      INFO      setup      starting manager
2025-12-24T13:40:05Z      INFO      controller-runtime.probe-api  starting probes server {"path": "/probes"}
2025-12-24T13:40:05Z      INFO      controller-runtime.metrics    Starting metrics server
2025-12-24T13:40:05Z      INFO      controller-runtime.metrics    Serving metrics server {"bindAddress": ":8080", "secure": false}
2025-12-24T13:40:05Z      INFO      starting server {"name": "health probe", "addr": "[::]:8081"}
I1224 13:40:05.949435      1 leaderelection.go:257] attempting to acquire leader lease kubedns-system/190f3354.kubedns.io...
I1224 13:40:21.476431      1 leaderelection.go:271] successfully acquired lease kubedns-system/190f3354.kubedns.io
2025-12-24T13:40:21Z      INFO      Starting EventSource {"controller": "kubedns", "controllerGroup": "security.kubedns.io", "controllerKind": "Kubedns", "source": "kind source: *v1.Kubedns"}
2025-12-24T13:40:21Z      DEBUG     events kubedns-controller-manager-589d4c6d6c-g44ld_32f847ed-6905-49b4-8803-205f1889e307 became leader {"type": "Normal", "object": {"kind": "Lease", "namespace": "kubedns-system", "name": "190f3354.kubedns.io", "uid": "a610f417-1a51-4948-a630-b85a8bc24c48", "apiVersion": "coordination.k8s.io/v1", "resourceVersion": "33521"}, "reason": "LeaderElection"}
2025-12-24T13:40:21Z      INFO      Starting Controller {"controller": "kubedns", "controllerGroup": "security.kubedns.io", "controllerKind": "Kubedns"}
2025-12-24T13:40:21Z      INFO      Starting workers {"controller": "kubedns", "controllerGroup": "security.kubedns.io", "controllerKind": "Kubedns", "worker count": 1}
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    Setting up the event listener...
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent adservice
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent cartservice
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent checkoutservice
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent currencyservice
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent emailservice
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent frontend
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent frontend-external
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent paymentservice
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent productcatalogservice
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent recommendationservice
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent redis-cart
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddServiceEvent shippingservice
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddPodEvent adservice-56d6f458b4-n46f8
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddPodEvent cartservice-7cb86d84f5-zpgs8
2025-12-24T13:40:21Z      INFO      kubedns.podEvents    EventHandler::AddPodEvent checkoutservice-7d7bb99d78-4cbvv
2025-12-24T13:40:30Z      INFO      kubedns.podEvents    EventHandler::AddPodEvent currencyservice-5f49c7988-qwnl9
2025-12-24T13:40:49Z      INFO      Recursive probing    Probe queue size: 22
2025-12-24T13:40:49Z      INFO      kubedns.podEvents    EventHandler::AddPodEvent emailservice-68cd6548d4-ws9hs
2025-12-24T13:40:51Z      INFO      kubedns.podEvents    Active pods: [checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9 adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8]
2025-12-24T13:40:51Z      INFO      kubedns.podEvents    Number of probes in State: 88
2025-12-24T13:41:11Z      INFO      Recursive probing    Probe queue size: 0
2025-12-24T13:41:11Z      INFO      Recursive probing    Running all probes again
2025-12-24T13:41:11Z      INFO      kubedns.podEvents    EventHandler::AddPodEvent frontend-56db6cc6df-7wqrt
2025-12-24T13:41:23Z      INFO      kubedns.podEvents    Active pods: [currencyservice-5f49c7988-qwnl9 emailservice-68cd6548d4-ws9hs adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
```

Dans la version 2, on autorise les communications frontend → checkout :

```
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl describe networkpolicy allow-frontend-to-checkout
Name:          allow-frontend-to-checkout
Namespace:     default
Created on:    2025-12-24 15:42:54 +0100 CET
Labels:        <none>
Annotations:   <none>
Spec:
  PodSelector:  app=checkoutservice
  Allowing ingress traffic:
    To Port: 5050/TCP
    From:
      PodSelector: app=frontend
  Not affecting egress traffic
  Policy Types: Ingress
```

```

user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl apply -f v2-frontend-checkout/allow-frontend-checkout.yaml
networkpolicy.networking.k8s.io/allow-frontend-to-checkout created
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl get networkpolicy
NAME                                POD-SELECTOR                      AGE
allow-frontend-to-checkout          app=checkoutservice              39s
deny-all                           <none>                           47m

```

Début des logs que nous obtenons :

```

user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl logs -n kubensonde-system deploy/kubensonde-controller-manager
2025-12-24T14:09:06Z INFO setup starting Kubensonde API server
2025-12-24T14:09:06Z INFO setup starting manager
2025-12-24T14:09:06Z INFO controller-runtime.probe-api starting probes server {"path": "/probes"}
2025-12-24T14:09:06Z INFO controller-runtime.metrics Starting metrics server
2025-12-24T14:09:06Z INFO controller-runtime.metrics Serving metrics server {"bindAddress": ":8080", "secure": false}
2025-12-24T14:09:06Z INFO starting server {"name": "health probe", "addr": "[:]:8081"}
I1224 14:09:06.683975 1 leaderelection.go:257] attempting to acquire leader lease kubensonde-system/190f3354.kubensonde.io...
I1224 14:09:23.460663 1 leaderelection.go:271] successfully acquired lease kubensonde-system/190f3354.kubensonde.io
2025-12-24T14:09:23Z DEBUG events kubensonde-controller-manager-589d4c6d6c-g44ld 895138ff-052f-4c7a-b8d7-3d074ad9dc61 became leader {"type": "Normal", "object": {"kind": "Lease", "namespace": "kubensonde-system", "name": "190f3354.kubensonde.io", "uid": "a610f417-1a51-4948-a630-b85a8bc24c48", "apiVersion": "coordination.k8s.io/v1", "resourceVersion": "35795"}, "reason": "LeaderElection"}
2025-12-24T14:09:23Z INFO Starting EventSource {"controller": "kubensonde", "controllerGroup": "security.kubensonde.io", "controllerKind": "Kubensonde", "source": "kind source: *v1.Kubensonde"}
2025-12-24T14:09:23Z INFO Starting Controller {"controller": "kubensonde", "controllerGroup": "security.kubensonde.io", "controllerKind": "Kubensonde"}
2025-12-24T14:09:23Z INFO Starting workers {"controller": "kubensonde", "controllerGroup": "security.kubensonde.io", "controllerKind": "Kubensonde", "worker count": 1}
2025-12-24T14:09:23Z INFO kubensonde.podEvents Setting up the event listener...
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent adservice
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent cartservice
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent checkoutservice
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent currencyservice
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent emailservice
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent frontend
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent frontend-external
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent paymentservice
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent productcatalogservice
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent recommendationservice
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent redis-cart
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddServiceEvent shippingservice
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddPodEvent adservice-56d6f458b4-n46f8
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddPodEvent cartservice-7cb86d84f5-zpgs8
2025-12-24T14:09:23Z INFO kubensonde.podEvents EventHandler::AddPodEvent checkoutservice-7d7bb99d78-4cbvv
2025-12-24T14:09:50Z INFO Kubensonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T14:09:53Z INFO kubensonde.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8]
2025-12-24T14:09:53Z INFO kubensonde.podEvents Number of probes in State: 43
2025-12-24T14:10:12Z INFO Kubensonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T14:10:20Z INFO Kubensonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T14:10:20Z INFO Kubensonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T14:10:20Z INFO Recursive probing Probe queue size: 0
2025-12-24T14:10:20Z INFO Recursive probing Running all probes again
2025-12-24T14:10:21Z INFO Kubensonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T14:10:21Z INFO kubensonde.podEvents EventHandler::AddPodEvent currencyservice-5f49c7988-qwnl9
2025-12-24T14:10:23Z INFO kubensonde.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T14:10:23Z INFO kubensonde.podEvents Number of probes in State: 58
2025-12-24T14:10:52Z INFO Kubensonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T14:10:53Z INFO kubensonde.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T14:10:53Z INFO kubensonde.podEvents Number of probes in State: 70

```

Dans la version 3, est autorisé l'égress du checkoutservice. Checkoutservice → services autorisés (vers les autres services c'est refusé grâce à la politique v1) :

```
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl apply -f v3-checkout-egress/allow-checkout-egress.yaml
networkpolicy.networking.k8s.io/checkout-egress created
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl get networkpolicy
NAME                                POD-SELECTOR  AGE
allow-frontend-to-checkout          app=checkoutservice  21m
checkout-egress                     app=checkoutservice  20s
deny-all                            <none>           68m
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl describe networkpolicy
allow-frontend-to-checkout  checkout-egress  deny-all
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl describe networkpolicy checkout-egress
Name:          checkout-egress
Namespace:     default
Created on:    2025-12-24 16:03:45 +0100 CET
Labels:        <none>
Annotations:   <none>
Spec:
  PodSelector:  app=checkoutservice
  Not affecting ingress traffic
  Allowing egress traffic:
    To Port: 7070/TCP
    To:
      PodSelector: app=cartservice
    -----
    To Port: 5000/TCP
    To:
      PodSelector: app=emailservice
    -----
    To Port: 50051/TCP
    To:
      PodSelector: app=paymentservice
    -----
    To Port: 50051/TCP
    To:
      PodSelector: app=shippingservice
    -----
    To Port: 3550/TCP
    To:
      PodSelector: app=productcatalogservice
    -----
    To Port: 7000/TCP
    To:
      PodSelector: app=currencyservice
  Policy Types: Egress
```


On sait que le checkoutservice peut uniquement sortir vers cartservice (7070/TCP), emailservice (5000/TCP), paymentsservice et shippingservice (50051/TCP), productcatalogservice (3550/TCP) et currencyservice (7000/TCP).

Une partie des logs :

```
6f458b4-n46f8 checkoutservice-7d7bb99d78-4cbvv frontend-56db6cc6df-7wqrt loadgenerator-59ccddd5c-5rnq9 recommendationservice-5856f9b4bb-5tr9s]
2025-12-24T15:13:07Z INFO kubsonde.podEvents Number of probes in State: 422
2025-12-24T15:13:07Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:08Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:08Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:08Z INFO Recursive probing Probe queue size: 0
2025-12-24T15:13:08Z INFO Recursive probing Running all probes again
2025-12-24T15:13:09Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:11Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:12Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:13Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:15Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:16Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:17Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:17Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:18Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:18Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:24Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:24Z INFO kubsonde.podEvents EventHandler::AddPodEvent shippingservice-6d787df574-ggjd5
2025-12-24T15:13:33Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:33Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:34Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:34Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:35Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:35Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:37Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:37Z INFO kubsonde.podEvents Active pods: [frontend-56db6cc6df-7wqrt loadgenerator-59ccddd5c-5rnq9 recommendationservice-5856f9b4bb-5tr9s redis-cart-c8ff86559-7vwq5 cartservice-7cb86d84f5-zpgs8 currencyservice-5f49c79
88-qwnl9 emailservice-68cd6548d4-ws9hs paymentsservice-845fc4d44-pd2mq productcatalogservice-6c6c594f5d-c9sv4 adservice-56d
6f458b4-n46f8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:13:37Z INFO kubsonde.podEvents Number of probes in State: 465
2025-12-24T15:13:37Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:38Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:38Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:39Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:39Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:40Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:41Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:41Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:42Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:43Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:44Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:44Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:45Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:47Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:49Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:13:49Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:14:07Z INFO kubsonde.podEvents Active pods: [cartservice-7cb86d84f5-zpgs8 currencyservice-5f49c79
88-qwnl9 emailservice-68cd6548d4-ws9hs paymentsservice-845fc4d44-pd2mq productcatalogservice-6c6c594f5d-c9sv4 adservice-56d
6f458b4-n46f8 checkoutservice-7d7bb99d78-4cbvv frontend-56db6cc6df-7wqrt loadgenerator-59ccddd5c-5rnq9 recommendationservice-5856f9b4bb-5tr9s redis-cart-c8ff86559-7vwq5]
2025-12-24T15:14:07Z INFO kubsonde.podEvents Number of probes in State: 485
2025-12-24T15:14:07Z INFO Kubsonde Runner Error when Probing: command terminated with exit code 1
user@deb:~/Documents/exam/microservices-demo/network-policies$
```

Dans la version 4 finale (en ayant appliqué les politiques réseau v1 à v3 auparavant), on autorise désormais l'egress vers redis-cart et l'ingress depuis cartservice :

```
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl apply -f v4-cart-redis/
networkpolicy.networking.k8s.io/cart-egress-redis created
networkpolicy.networking.k8s.io/redis-ingress-cart-only created
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl get networkpolicy
NAME                                POD-SELECTOR  AGE
allow-frontend-to-checkout         app=checkoutservice  42m
cart-egress-redis                  app=cartservice      9s
checkout-egress                   app=checkoutservice  21m
deny-all                          <none>              89m
redis-ingress-cart-only            app=redis-cart       9s
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl describe networkpolicy
allow-frontend-to-checkout  checkout-egress  redis-ingress-cart-only
cart-egress-redis          deny-all
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl describe networkpolicy cart-egress-redis
Name:      cart-egress-redis
Namespace: default
Created on: 2025-12-24 16:25:35 +0100 CET
Labels:    <none>
Annotations: <none>
Spec:
  PodSelector:  app=cartservice
  Not affecting ingress traffic
  Allowing egress traffic:
    To Port: 6379/TCP
    To:
      PodSelector: app=redis-cart
  Policy Types: Egress
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl describe networkpolicy redis-ingress-cart-only
Name:      redis-ingress-cart-only
Namespace: default
Created on: 2025-12-24 16:25:35 +0100 CET
Labels:    <none>
Annotations: <none>
Spec:
  PodSelector:  app=redis-cart
  Allowing ingress traffic:
    To Port: 6379/TCP
    From:
      PodSelector: app=cartservice
  Not affecting egress traffic
  Policy Types: Ingress
user@deb:~/Documents/exam/microservices-demo/network-policies$
```

Maintenant :

- Le flux frontend → checkoutservice est autorisé par la NetworkPolicy "allow-frontend-to-checkout".
- Le flux frontend → redis-cart est refusé par la NetworkPolicy "deny-all".
- Le flux cartservice → redis-cart est autorisé par les NetworkPolicies "cart-egress-redis" et "redis-ingress-cart-only".
- Le flux cartservice → paymentsservice est refusé par la NetworkPolicy "deny-all".
- La NetworkPolicy "checkout-egress" autorise checkoutservice à envoyer des données vers cartservice, emailservice, paymentsservice, shippingservice, productcatalogservice et currencyservice.

Logs :

```
user@deb:~/Documents/exam/microservices-demo/network-policies$ kubectl logs -n kubernetes-system deploy/kubernetes-controller-manager
2025-12-24T15:44:25Z INFO setup starting Kubernetes API server
2025-12-24T15:44:25Z INFO setup starting manager
2025-12-24T15:44:25Z INFO controller-runtime.probe.api starting probes server {"path": "/probes"}
2025-12-24T15:44:25Z INFO controller-runtime.metrics starting metrics server
11224 15:44:25.951934 I leaderelection.go:257] attempting to acquire leader lease kubernetes-system/190f3354.kubernetes.io...
2025-12-24T15:44:25Z INFO starting server {"name": "health probe", "addr": "[::]:8081"}
2025-12-24T15:44:25Z INFO controller-runtime.metrics serving metrics server {"bindAddress": ":8080", "secure": false}
11224 15:44:42.169865 I leaderelection.go:271] successfully acquired lease kubernetes-system/190f3354.kubernetes.io
2025-12-24T15:44:42Z DEBUG events kubernetes-controller-manager-589d4c6d6c-g44ld_314ca572-5f78-4905-96c7-fbd1fa0b9988 became leader {"type": "Normal", "object": {"kind": "Lease", "namespace": "kubernetes-system", "name": "190f3354.kubernetes.io", "uid": "a618f417-1a51-4948-a038-b85a8bc24c48", "apiVersion": "coordination.k8s.io/v1", "resourceVersion": "43248"}, "reason": "LeaderElection"}
2025-12-24T15:44:42Z INFO Starting Controller {"controller": "Kubernetes", "controllerGroup": "security.kubernetes.io", "controllerKind": "Kubernetes", "source": "kind source: *v1.Kubernetes"}
2025-12-24T15:44:42Z INFO Starting workers {"controller": "Kubernetes", "controllerGroup": "security.kubernetes.io", "controllerKind": "Kubernetes", "worker count": 1}
2025-12-24T15:44:42Z INFO kubernetes.podEvents Setting up the event listener...
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent adservice
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent cartservice
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent checkoutservice
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent currencyservice
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent emailservice
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent frontend
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent frontend-external
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent paymentservice
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent productcatalogservice
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent recommendationservice
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent redis-cart
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddServiceEvent shippingservice
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddPodEvent adservice-56d6f458b4-n46f8
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddPodEvent cartservice-7cb86d84f5-zpgs8
2025-12-24T15:44:42Z INFO kubernetes.podEvents EventHandler: AddPodEvent checkoutservice-7d7bb99d78-4cbvv
2025-12-24T15:45:11Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:45:12Z INFO Kubernetes Runner Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 adservice-56d6f458b4-n46f8]
2025-12-24T15:45:12Z INFO Kubernetes Runner Number of probes in State: 43
2025-12-24T15:45:31Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:45:31Z INFO Kubernetes Runner EventHandler: AddPodEvent currencyservice-5f49c7988-qwnl9
2025-12-24T15:45:42Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:46:07Z INFO kubernetes.podEvents Number of probes in State: 56
2025-12-24T15:46:07Z INFO kubernetes.podEvents Error when Probing: command terminated with exit code 1
2025-12-24T15:46:12Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:46:14Z INFO kubernetes.podEvents Number of probes in State: 68
2025-12-24T15:46:29Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:46:44Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:46:44Z INFO kubernetes.podEvents Number of probes in State: 68
2025-12-24T15:47:03Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:47:13Z INFO Recursive probing Probe queue size: 0
2025-12-24T15:47:13Z INFO Recursive probing Running all probes again
2025-12-24T15:47:14Z INFO kubernetes.podEvents Error when Probing: command terminated with exit code 1
2025-12-24T15:47:14Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:47:14Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:47:47Z INFO kubernetes.podEvents Active pods: [cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv adservice-56d6f458b4-n46f8]
2025-12-24T15:47:47Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:48:16Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
-----
2025-12-24T15:48:19Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:48:19Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:48:36Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:48:49Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:48:49Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:49:03Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:49:19Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:49:19Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:49:50Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:49:50Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:50:12Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:50:51Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:50:51Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:51:12Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:51:21Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:51:21Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:51:23Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:51:35Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:51:51Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:51:51Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:52:22Z INFO kubernetes.podEvents Active pods: [cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv adservice-56d6f458b4-n46f8]
2025-12-24T15:52:22Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:52:30Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:52:36Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:52:37Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:52:53Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:52:53Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:53:24Z INFO kubernetes.podEvents Active pods: [checkoutservice-7d7bb99d78-4cbvv adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8]
2025-12-24T15:53:24Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:53:49Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:53:56Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:53:56Z INFO kubernetes.podEvents Number of probes in State: 70
2025-12-24T15:54:20Z INFO kubernetes.podEvents EventHandler: AddPodEvent emailservice-68cd6548d4-ws9hs
2025-12-24T15:54:27Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9]
2025-12-24T15:54:27Z INFO kubernetes.podEvents Number of probes in State: 94
2025-12-24T15:54:57Z INFO kubernetes.podEvents Error when Probing: command terminated with exit code 1
2025-12-24T15:54:57Z INFO kubernetes.podEvents Active pods: [checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9 adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8]
2025-12-24T15:54:57Z INFO kubernetes.podEvents Number of probes in State: 97
2025-12-24T15:55:08Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:55:29Z INFO kubernetes.podEvents Active pods: [cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9 adservice-56d6f458b4-n46f8]
2025-12-24T15:55:29Z INFO kubernetes.podEvents Number of probes in State: 97
2025-12-24T15:55:46Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:55:59Z INFO kubernetes.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9]
2025-12-24T15:55:59Z INFO kubernetes.podEvents Number of probes in State: 97
2025-12-24T15:56:22Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:30Z INFO kubernetes.podEvents Active pods: [cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9 adservice-56d6f458b4-n46f8]
2025-12-24T15:56:30Z INFO kubernetes.podEvents Number of probes in State: 97
2025-12-24T15:56:31Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:31Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:32Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:32Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:34Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:34Z INFO Kubernetes Runner Error when Probing: command terminated with exit code 1
```

```

2025-12-24T15:56:30Z INFO kubesonde.podEvents Number of probes in State: 97
2025-12-24T15:56:31Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:31Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:31Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:32Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:34Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:34Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:36Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:37Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:38Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:43Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:44Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:44Z INFO Recursive probing Probe queue size: 0
2025-12-24T15:56:44Z INFO Recursive probing Running all probes again
2025-12-24T15:56:44Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:44Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:45Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:56:45Z INFO Kubesonde.podEvents EventHandler: AddPodEvent frontend-56db6cc6df-7wqrt
2025-12-24T15:56:45Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:57:00Z INFO kubesonde.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9 emailservice-68cd6548d4-ws9hs]
2025-12-24T15:57:00Z INFO kubesonde.podEvents Number of probes in State: 149
2025-12-24T15:57:20Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:57:32Z INFO kubesonde.podEvents Active pods: [cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9 emailservice-68cd6548d4-ws9hs adservice-56d6f458b4-n46f8]
2025-12-24T15:57:32Z INFO kubesonde.podEvents Number of probes in State: 149
2025-12-24T15:57:56Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:58:02Z INFO kubesonde.podEvents Active pods: [currencyservice-5f49c7988-qwnl9 emailservice-68cd6548d4-ws9hs adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv]
2025-12-24T15:58:02Z INFO kubesonde.podEvents Number of probes in State: 149
2025-12-24T15:58:14Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:58:30Z INFO Kubesonde Runner Error when Probing: command terminated with exit code 1
2025-12-24T15:58:30Z INFO Kubesonde.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9 emailservice-68cd6548d4-ws9hs]
2025-12-24T15:58:32Z INFO kubesonde.podEvents Active pods: [adservice-56d6f458b4-n46f8 cartservice-7cb86d84f5-zpgs8 checkoutservice-7d7bb99d78-4cbvv currencyservice-5f49c7988-qwnl9 emailservice-68cd6548d4-ws9hs]
2025-12-24T15:58:32Z INFO kubesonde.podEvents Number of probes in State: 149

```

En conclusion, nous avons introduit progressivement ensuite des NetworkPolicies, en commençant par une politique restrictive (deny all), puis en autorisant uniquement les communications requises, étape par étape, pour observer les flux via KubeSonde et valider ou non la connectivité après chaque itération. A chaque échec de connectivité, nous avons rajouté une NetworkPolicy minimale autorisant uniquement le flux demandé. L'objectif de cette approche a été d'accorder aussi peu de privilèges que possible pour ne pas oublier de sécuriser l'accès à des ressources involontairement et ainsi protéger au mieux la boutique.